Linear equations in one variable — Part 1 answer key

#	ID	Answer	Difficulty
1.1	85789084-6a50-4310-ae89-d2067332bac0	C	Easy
1.2	a7ee5a49-de3b-4c5b-a46e-d593e051326d	C	Easy
1.3	cb01072a-91f8-4bc2-b74f-51d996629f69	C	Easy
1.4	96677815-bb45-424b-a4e5-ec40dbe43bb8	B	Easy
1.5	67336474-3a8f-4702-9694-98a3abdcc31a	.2, 1/5	Easy
1.6	62a3416c-5c61-4965-82b5-d0803a5d0182	27	Easy
1.7	5b069676-b4f1-4f88-aa86-55183f3f0055	A	Easy
1.8	d4e10666-e119-46af-8dba-a8d527aeca75	D	Easy
1.9	26011899-7805-4655-b434-d6a03ef7d8de	480	Easy
1.10	fe2efe64-2759-4c9c-b6ac-4e4f9f1c9372	D	Easy
1.11	022296-DC	?	Easy
1.12	214d64ec-0e1a-4a3f-8fcb-73b3eecf4a6b	21	Easy
1.13	5b55a923-dce4-4bf9-a037-3f16263d1743	D	Easy
1.14	08682-DC	A	Easy
1.15	9931da93-9c67-4834-ac2f-f3587e875e3e	C	Easy
1.16	022240-DC	B	Easy
1.17	8076db82-ac6c-4ac2-a6b7-46a9a4b4059b	D	Easy
1.18	9d2c51d3-1801-4e1e-b2f2-9e58978c2f8e	C	Easy
1.19	14894-DC	C	Easy
1.20	8cb965d3-c0cb-4549-8b51-5decc372d910	A	Easy

1.21	d08114ae-ef2f-4e5b-a0ef-67d3bb996aa9	B	Easy
1.22	06902-DC	?	Easy
1.23	84445f9f-c91d-47ad-a7fb-f1adeb091390	C	Easy
1.24	bcd38cac-e6c1-411f-ae0b-50e4ae48adc1	A	Easy
1.25	8a9230ef-2284-4f11-9f4e-abff1fd55a4e	0.2, 1/5	Easy
1.26	7e8a4250-4838-4f9d-9e6c-24235e682f46	B	Easy
1.27	974b35b6-beab-44b9-8d36-7f18c4daa0d1	A	Easy
1.28	0f1aa9d1-12d4-4e22-b2aa-c753912a2a2e	A	Easy
1.29	c754edce-0af7-420c-8fc5-0e5f0cd80b73	34	Easy
1.30	9055c207-d1fc-4110-be3c-1ea08fcd52ab	6	Easy
1.31	c57e37c9-32b5-4879-b56c-ae8aec1a46fc	A	Easy
1.32	0aa2936b-88d9-4e8d-a776-486d5d1f93ba	40	Easy
1.33	c33a8ce0-debc-4709-b0d7-3f1800613186	8.6, 43/5	Easy
1.34	020349-DC	B	Easy
1.35	7ea2442b-48e4-447b-a75d-c1614b613123	B	Easy
1.36	ee16e929-f3b9-4883-9cb5-c95afc6fb0df	B	Easy
1.37	2ed3a51e-14c9-41a5-8246-b0df7d098ee8	35	Easy
1.38	05768-DC	D	Easy
1.39	ab4c9e76-1929-4d3f-ad64-4a63c8386545	D	Easy
1.40	07141-DC	A	Easy
1.41	208ae094-8f06-4329-9993-51169b92549f	B	Easy
1.42	2870e625-1e5c-4b4d-be09-4cc08fc8d83b	C	Easy
1.43	b80105ad-11a3-443a-9dc8-51472ba42bbf	10	Easy
1.44	87d05c33-deec-41ec-8d9c-f0a0094beae6	C	Easy
1.45	70f8f5ea-a016-4c7d-ad77-2b55a0b1c990	B	Easy
1.46	f815b832-89ea-4f49-bc05-5f5c25cdb59f	55	Easy

1.47	015193-DC	C	Easy
1.48	1da5e7f6-df74-4a25-9976-1f105af7942e	B	Easy
1.49	9df9861f-e14c-45b9-be5c-1d9bb6fbafba	54	Easy
1.50	f2ee68d6-91e0-4510-a278-ad0aa7e1153c	A	Easy
1.51	cab807ab-67ed-464f-a4b2-ece6edb417b6	8	Easy
1.52	5bbff16a-8cdc-4d4c-b45c-545cf12538e2	B	Easy
1.53	151077e0-2b11-4c07-bc61-40cb9fc124a8	B	Easy
1.54	10931-DC	D	Easy
1.55	5839b7ae-3235-44a9-9666-5735f17e34d8	A	Easy
1.56	a3e37333-8679-4d42-b7c8-1b15541bd74d	C	Easy

Rationales

▼ 1.1 — C

Choice C is correct. Dividing all terms in the given equation by 4 yields $\frac{4x}{4} - \frac{28}{4} = -\frac{24}{4}$, or $x - 7 = -6$. Therefore, the value of $x - 7$ is -6.

Choice A is incorrect. This is the value of $4x - 28$, not $x - 7$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.2 — C

Choice C is correct. It's given that $x = 7$. Substituting 7 for x into the given expression $x + 20$ yields $7 + 20$, which is equivalent to 27.

Choice A is incorrect. This is the value of $x + 6$.

Choice B is incorrect. This is the value of $x + 13$.

Choice D is incorrect. This is the value of $x + 27$.

▼ 1.3 — C

Choice C is correct. It's given that $4x = 3$. Multiplying each side of this equation by 6 yields $24x = 18$. Therefore, the value of $24x$ is 18.

Choice A is incorrect. This is the value of $6x$, not $24x$.

Choice B is incorrect. This is the value of $8x$, not $24x$.

Choice D is incorrect. This is the value of $96x$, not $24x$.

▼ 1.4 — B

Choice B is correct. It's given that the film club has 90 members on the first day of a semester, and 10 new members join the film club each day after the first day of the semester. This means that after 4 days, 4×10, or 40, new members will have joined the club. Adding 40 members to the original 90 club members yields 130 members. Thus, the film club will have 130 total members 4 days after the first day of the semester.

Choice A is incorrect. This is the number of members that will have joined the film club 4 days after the first day of the semester if 100 new members, not 10, join the film club each day.

Choice C is incorrect. This is the number of members the film club will have 4 days after the first day of the semester if 1 new member, not 10, joins the film club each day.

Choice D is incorrect. This is the number of members the film club has on the first day of the semester.

▼ 1.5 — .2, 1/5

The correct answer is $\frac{1}{5}$. Since the number 5 can also be written as $\frac{5}{1}$, the given equation can also be written as $\frac{x}{8} = \frac{5}{1}$. This equation is equivalent to $\frac{8}{x} = \frac{1}{5}$. Therefore, the value of $\frac{8}{x}$ is $\frac{1}{5}$. Note that 1/5 and .2 are examples of ways to enter a correct answer.

Alternate approach: Multiplying both sides of the equation $\frac{x}{8} = 5$ by 8 yields $x = 40$. Substituting 40 for x into the expression $\frac{8}{x}$ yields $\frac{8}{40}$, or $\frac{1}{5}$.

▼ 1.6 — 27

The correct answer is 27. Multiplying both sides of the given equation by 3 yields $36 + x = 39$, or $18 + 3x = 27$. Therefore, the value of $18 + 3x$ is 27.

▼ 1.7 — A

Choice A is correct. It's given that a principal used a total of 25 blue flags and yellow flags. It's also given that of the 25 flags used, 20 flags were blue. Subtracting the number of blue flags used from the total number of flags used results in the number of yellow flags used. It follows that the number of yellow flags used is $25 - 20$, or 5.

Choice B is incorrect. This is the number of blue flags used.

Choice C is incorrect. This is the total number of flags used.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.8 — D

Choice D is correct. The given phrase "8 times a number x" can be represented by the expression $8x$. The given phrase "3 more than" indicates an increase of 3 to a quantity. Therefore "3 more than 8 times a number x" can be represented by the expression $8x + 3$. Since it's given that 3 more than 8 times a number x is equal to 83, it follows that $8x + 3$ is equal to 83, or $8x + 3 = 83$. Therefore, the equation that represents this situation is $8x + 3 = 83$.

Choice A is incorrect. This equation represents 3 times the quantity 8 times a number x is equal to 83.

Choice B is incorrect. This equation represents 8 times a number x is equal to 3 more than 83.

Choice C is incorrect. This equation represents 8 more than 3 times a number x is equal to 83.

▼ 1.9 — 480

The correct answer is 480. Multiplying both sides of the given equation by 8 yields $82 + x = 860$, or $16 + 8x = 480$. Therefore, if $2 + x = 60$, the value of $16 + 8x$ is 480.

▼ 1.10 — D

Choice D is correct. Subtracting $16x$ from both sides of the given equation yields $24 = 8x$. Dividing both sides of this equation by 8 yields $3 = x$. Therefore, the solution to the given equation is 3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.11 — ?

The correct answer is 24. Multiplying both sides of the given equation by 3 yields $3(3x+2)=24$. Using the distributive property to rewrite the left-hand side of this equation yields $9x+6=24$.

▼ 1.12 — 21

The correct answer is 21. It's given that the equation $3x + 21 = 3x + k$ has infinitely many solutions. If an equation in one variable has infinitely many solutions, then the equation is true for any value of the variable. Subtracting $3x$ from both sides of the given equation yields $k = 21$. Since this equation must be true for any value of x, the value of k is 21.

▼ 1.13 — D

Choice D is correct. Multiplying both sides of the given equation by 10 yields $10(9x + 4) = 10(67)$, or $90x + 40 = 670$. Therefore, the value of $90x + 40$ is 670.

Choice A is incorrect. This is the value of x, not $90x + 40$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.14 — A

Choice A is correct. Multiplying both sides of the equation by 5 results in $4x=100$. Dividing both sides of the resulting equation by 4 results in $x=25$.

Choice B is incorrect and may result from adding 20 and 4. Choice C is incorrect and may result from dividing 20 by 5 and then multiplying the result by 4. Choice D is incorrect and may result from subtracting 5 from 20.

▼ 1.15 — C

Choice C is correct. It's given that $x = 40$. Adding 6 to both sides of this equation yields $x + 6 = 40 + 6$, or $x + 6 = 46$. Therefore, the value of $x + 6$ is 46.

Choice A is incorrect. This is the value of $x - 6$, not $x + 6$.

Choice B is incorrect. This is the value of x, not $x + 6$.

Choice D is incorrect. This is the value of $x + 24$, not $x + 6$.

▼ 1.16 — B

Choice B is correct. Multiplying both sides of the given equation by 5 yields $2n = 50$. Substituting 50 for $2n$ in the expression $2n - 1$ yields $50 - 1 = 49$.

Alternate approach: Dividing both sides of $2n = 50$ by 2 yields $n = 25$. Evaluating the expression $2n - 1$ for $n = 25$ yields $2(25) - 1 = 49$.

Choice A is incorrect and may result from finding the value of $n - 1$ instead of $2n - 1$. Choice C is incorrect and may result from finding the value of $2n$ instead of $2n - 1$. Choice D is incorrect and may result from finding the value of $4n - 1$ instead of $2n - 1$.

▼ 1.17 — D

Choice D is correct. It's given that $3x - 8 = 7$. Adding 8 to both sides of this equation yields $3x = 15$. Adding 8 to both sides of this equation yields $3x + 8 = 23$. Therefore, the value of $3x + 8$ is 23.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x, not $3x + 8$.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.18 — C

Choice C is correct. It's given that the perimeter of an isosceles triangle is 36 feet and that each of the two congruent sides has a length of 10 feet. The perimeter of a triangle is the sum of the lengths of its three sides. The equation $10 + 10 + x = 36$ can be used to represent this situation, where x is the length, in feet, of the third side. Combining like terms on the

left-hand side of this equation yields $20 + x = 36$. Subtracting 20 from each side of this equation yields $x = 16$. Therefore, the length, in feet, of the third side is 16.

Choice A is incorrect. This would be the length, in feet, of the third side if the perimeter was 30 feet, not 36 feet.

Choice B is incorrect. This would be the length, in feet, of the third side if the perimeter was 32 feet, not 36 feet.

Choice D is incorrect. This would be the length, in feet, of the third side if the perimeter was 38 feet, not 36 feet.

▼ 1.19 — C

Choice C is correct. To make it easier to add like terms on the left-hand side of the given equation, both sides of the equation can be multiplied by 6, which is the lowest common denominator of $\frac{1}{2}$ and $\frac{1}{6}$. This yields $3x - x = 6$, which can be rewritten as $2x = 6$. Dividing both sides of this equation by 2 yields $x = 3$.

Choice A is incorrect and may result from subtracting the denominators instead of numerators with common denominators to get $-\frac{1}{4}x$, rather than $\frac{1}{3}x$, on the left-hand side of the equation. Choice B is incorrect and may result from rewriting the given equation as $\frac{1}{2}x = \frac{1}{6}$ instead of $2x = 6$. Choice D is incorrect and may result from conceptual or computational errors.

▼ 1.20 — A

Choice A is correct. It's given that a total of 165 people contributed to a charity event as either a donor or a volunteer. It's also given that 130 people contributed as a donor. It follows that $165 - 130$, or 35, people contributed as a volunteer.

Choice B is incorrect. This is the number of people who contributed as a donor, not a volunteer.

Choice C is incorrect. This is the total number of people who contributed as either a donor or a volunteer, not the number of people who contributed as a volunteer.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.21 — B

Choice B is correct. Subtracting 12 from each side of the given equation yields $3x - 12 = 30 - 12$, or $3x - 12 = 18$. Therefore, the value of $3x - 12$ is 18.

Choice A is incorrect. This is the value of $x - 12$, not $3x - 12$.

Choice C is incorrect. This is the value of $x + 12$, not $3x - 12$.

Choice D is incorrect. This is the value of $3x + 12$, not $3x - 12$.

▼ 1.22 — ?

Choice B is correct. Subtracting the number of books left over from the total number of books results in $43-7=36$, which is the number of books distributed. Dividing the number of books distributed by the number of books given to each child results in $\frac{36}{2}=18$.

Choice A is incorrect and results from dividing the total number of books by the number of books given to each child, $\frac{43}{2}\approx22$, then subtracting the number of books left over from the result, $22-7=15$. Choice C is incorrect and results from adding the number of books left over to the total number of books, $43+7=50$, then dividing the result by the number of books given to each child, $\frac{50}{2}=25$. Choice D is incorrect and results from dividing the total number of books by the number of books given to each child, $\frac{43}{2}\approx22$, then adding the number of books left over, $22+7=29$.

▼ 1.23 — C

Choice C is correct. It's given that $8x = 6$. Multiplying each side of this equation by 9 yields $72x = 54$. Therefore, the value of $72x$ is 54.

Choice A is incorrect. This is the value of $4x$, not $72x$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.24 — A

Choice A is correct. Subtracting 8 from both sides of the given equation yields $p + 3 = 2$. Subtracting 3 from both sides of this equation yields $p = -1$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.25 — 0.2, 1/5

The correct answer is .2 . Subtracting 2.6 from each side of the given equation yields $x = 0.2$. Therefore, the value of x that's the solution to the given equation is 0.2. Note that .2 and 1/5 are examples of ways to enter a correct answer.

▼ 1.26 — B

Choice B is correct. It's given that the shop's inventory starts with $4{,}500$ paper cups and that the manager estimates that 70 of these paper cups are used each day. Let x represent the number of days in which the estimated supply of paper cups will reach $1{,}700$. The equation $4{,}500 - 70x = 1{,}700$ represents this situation. Subtracting $4{,}500$ from both sides of this equation yields $-70x = -2{,}800$. Dividing both sides of this equation by -70 yields $x = 40$. Therefore, based on this estimate, the supply of paper cups will reach $1{,}700$ in 40 days.

Choice A is incorrect. After 20 days, the estimated supply of paper cups would be $4{,}500 - 70(20)$, or $3{,}100$ cups, not $1{,}700$ cups.

Choice C is incorrect. After 60 days, the estimated supply of paper cups would be $4{,}500 - 70(60)$, or 300 cups, not $1{,}700$ cups.

Choice D is incorrect. After 80 days, the estimated supply of paper cups would be $4{,}500 - 70(80)$, or $-1{,}100$ cups, which isn't possible.

▼ 1.27 — A

Choice A is correct. Applying the distributive property on the left-hand side of the given equation yields $3x + 5x + 20 = 76$, or $8x + 20 = 76$. Subtracting 20 from each side of this equation yields $8x = 56$. Dividing each side of this equation by 8 yields $x = 7$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the solution to the equation $x + 4 = 76$, not $3x + 5x + 4 = 76$.

▼ 1.28 — A

Choice A is correct. Dividing both sides of the given equation by 8 yields $x = 11$. Therefore, 11 is the solution to the given equation.

Choice B is incorrect. This is the solution to the equation $x + 8 = 88$.

Choice C is incorrect. This is the solution to the equation $x - 8 = 88$.

Choice D is incorrect. This is the solution to the equation $\frac{x}{8} = 88$.

▼ 1.29 — 34

The correct answer is 34. It's given that a line segment has a length of 115 cm and is divided into three parts, where one part is 47 cm long and the other two parts have lengths that are equal. If x represents the length, in cm, of each of the two parts of equal length, then the equation $47 + x + x = 115$, or $47 + 2x = 115$, represents this situation. Subtracting 47 from each side of this equation yields $2x = 68$. Dividing each side of this equation by 2 yields $x = 34$. Therefore, the length, in cm, of one of the two parts of equal length is 34.

▼ 1.30 — 6

The correct answer is 6. Dividing both sides of the equation $6n = 12$ by 6 yields $n = 2$. Substituting 2 for n in the expression $n + 4$ yields $2 + 4$, or 6.

▼ 1.31 — A

Choice A is correct. Subtracting 180 from both sides of the given equation yields $5p = 70$. Dividing both sides of this equation by 5 yields $p = 14$. Therefore, the value of p that satisfies the equation $5p + 180 = 250$ is 14.

Choice B is incorrect. This value of p satisfies the equation $5p + 180 = 505$.

Choice C is incorrect. This value of p satisfies the equation $5p + 180 = 610$.

Choice D is incorrect. This value of p satisfies the equation $5p + 180 = 1{,}430$.

▼ 1.32 — 40

The correct answer is 40. Subtracting 5 from both sides of the given equation yields $4x = 160$. Dividing both sides of this equation by 4 yields $x = 40$. Therefore, the solution to the given equation is 40.

▼ 1.33 — 8.6, 43/5

The correct answer is 8.6. Dividing both sides of the given equation by 10 yields $x = 8.6$. Therefore, 8.6 is the solution to the given equation. Note that 8.6 and 43/5 are examples of ways to enter a correct answer.

▼ 1.34 — B

Choice B is correct. Subtracting 4 from each side of the given equation yields $6 = 2x$, or $x = 3$, so the equation has a unique solution of $x = 3$.

Choice A is incorrect. Since 3 is a value of x that satisfies the given equation, the equation has at least 1 solution. Choice C is incorrect. Linear equations can have 0, 1, or infinitely many solutions; no linear equation has exactly 3 solutions. Choice D is incorrect. If a linear equation has infinitely many solutions, it can be reduced to $0 = 0$. This equation reduces to $x = 3$, so there is only 1 solution.

▼ 1.35 — B

Choice B is correct. Dividing each side of the given equation by 3 yields $x - 9 = 8$. Therefore, the value of $x - 9$ is 8.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $3x - 27$, not $x - 9$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.36 — B

Choice B is correct. For a restaurant owner to purchase a pizza oven at a total price of 13,000 dollars, a onetime down payment is required, and then fixed monthly payments are made for the remaining amount owed for the pizza oven. It's given that the equation $13{,}000 = 2{,}800 + 200t$ represents this situation, where t is the number of fixed monthly payments that are made. In this equation, 2,800 is a constant term, which represents the

onetime down payment, in dollars, and 200 is the coefficient of t. Multiplying 200 by t gives the total amount, in dollars, paid by the fixed monthly payments. Therefore, the best interpretation of 200 in this context is the amount, in dollars, of each fixed monthly payment.

Choice A is incorrect. This is the best interpretation of 2,800, not 200, in this context.

Choice C is incorrect. This is the best interpretation of 13,000, not 200, in this context.

Choice D is incorrect. This is the best interpretation of the variable t, not 200, in this context.

▼ 1.37 — 35

The correct answer is 35. It's given that the perimeter of an isosceles triangle is 83 inches and that each of the two congruent sides has a length of 24 inches. The perimeter of a triangle is the sum of the lengths of its three sides. The equation $24 + 24 + x = 83$ can be used to represent this situation, where x is the length, in inches, of the third side. Combining like terms on the left-hand side of this equation yields $48 + x = 83$. Subtracting 48 from both sides of this equation yields $x = 35$. Therefore, the length, in inches, of the third side is 35.

▼ 1.38 — D

Choice D is correct. Dividing each side of the original equation by 2 yields $\frac{4x+6}{2} = \frac{12}{2}$, which simplifies to $2x+3=6$.

Choice A is incorrect. Dividing each side of the original equation by 2 gives $2x+3=6$, which is not equivalent to $2x+4=6$. Choice B is incorrect. Dividing each side of the original equation by 4 gives $x+\frac{3}{2}=3$, which is not equivalent to $x+3=3$. Choice C is incorrect. Dividing each side of the original equation by 3 gives $\frac{4}{3}x+2=4$, which is not equivalent to $3x+2=4$.

▼ 1.39 — D

Choice D is correct. It's given that $5x = 20$. Multiplying both sides of this equation by 3 yields $15x = 60$. Therefore, the value of $15x$ is 60.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 1.40 — A

Choice A is correct. If one pound of grapes costs $2, two pounds of grapes will cost 2 times $2, three pounds of grapes will cost 3 times $2, and so on. Therefore, c pounds of grapes will cost c times $2, which is 2c dollars.

Choice B is incorrect and may result from incorrectly adding instead of multiplying. Choice C is incorrect and may result from assuming that c pounds cost $2, and then finding the cost per pound. Choice D is incorrect and could result from incorrectly assuming that 2 pounds cost $c, and then finding the cost per pound.

▼ 1.41 — B

Choice B is correct. Subtracting 7 from each side of the given equation yields $w = 350$. Therefore, the value of w that is the solution to the given equation is 350.

Choice A is incorrect. This is the value of w that is the solution to the equation $7w = 357$, not $w + 7 = 357$.

Choice C is incorrect. This is the value of w that is the solution to the equation $w - 7 = 357$, not $w + 7 = 357$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.42 — C

Choice C is correct. It's given that John made a $\$16$ payment each month for p months. The total amount of these payments can be represented by the expression $16p$. The down payment can be added to that amount to find the total amount John paid, yielding the expression $16p + 37$. It's given that John paid a total of $\$165$. Therefore, the expression for the total amount John paid can be set equal to that amount, yielding the equation $16p + 37 = 165$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.43 — 10

The correct answer is 10. It's given that the cost for the entire purchase was $\$27$ after a coupon was used for $\$63$ off the entire purchase. Adding the amount of the coupon to the purchase price yields $27 + 63 = 90$. Thus, the cost for the entire purchase before using the coupon was $\$90$. It's given that Nasir bought 9 storage bins. The original price for 1 storage bin can be found by dividing the total cost by 9. Therefore, the original price, in dollars, for 1 storage bin is $\frac{90}{9}$, or 10.

▼ 1.44 — C

Choice C is correct. It's given that a gym charges its members a onetime $\$36$ enrollment fee and a membership fee of $\$19$ per month. Let x represent the number of months at the gym after which a member will have been charged a total of $\$188$. If there are no charges other than the enrollment fee and the membership fee, the equation $36 + 19x = 188$ can be used to represent this situation. Subtracting 36 from both sides of this equation yields $19x = 152$. Dividing both sides of this equation by 19 yields $x = 8$. Therefore, a member will have been charged a total of $\$188$ after 8 months at the gym.

Choice A is incorrect. A member will have been charged a total of $\$36 + 19 \times 4$, or $\$112$, after 4 months at the gym.

Choice B is incorrect. A member will have been charged a total of $\$36 + 19 \times 5$, or $\$131$, after 5 months at the gym.

Choice D is incorrect. A member will have been charged a total of $\$36 + 19 \times 10$, or $\$226$, after 10 months at the gym.

▼ 1.45 — B

Choice B is correct. Subtracting 275 from both sides of the given equation yields $2p = 50$. Dividing both sides of this equation by 2 yields $p = 25$. Therefore, the value of p that satisfies the given equation is 25.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of p that satisfies the equation $2 + p + 275 = 325$, not $2p + 275 = 325$.

Choice D is incorrect. This is the value of p that satisfies the equation $2p - 275 = 325$, not $2p + 275 = 325$.

▼ 1.46 — 55

The correct answer is 55. Subtracting 40 from both sides of the given equation yields $x = 55$. Therefore, the value of x is 55.

▼ 1.47 — C

Choice C is correct. Subtracting 3 from both sides of the given equation yields $2x = 4$. Dividing both sides by 2 results in $x = 2$.

Choices A and B are incorrect and may result from computational errors. Choice D is incorrect. This is the value of $2x$.

▼ 1.48 — B

Choice B is correct. Dividing both sides of the given equation $7x = 28$ by 7 yields $x = 4$. Substituting 4 for x in the expression $8x$ yields 84, which is equivalent to 32.

Choice A is incorrect. This is the value of $\frac{21}{4}x$.

Choice C is incorrect. This is the value of $42x$.

Choice D is incorrect. This is the value of $56x$.

▼ 1.49 — 54

The correct answer is 54. Dividing both sides of the given equation by 2 yields $x = 6$. Multiplying both sides of this equation by 9 yields $9x = 54$. Thus, the value of $9x$ is 54.

▼ 1.50 — A

Choice A is correct. Dividing each side of the given equation by 7 yields $\frac{72x - 3}{7} = \frac{63}{7}$, or $2x - 3 = 9$. Therefore, the equation $2x - 3 = 9$ is equivalent to the given equation and has the same solution.

Choice B is incorrect. This equation is equivalent to $72x - 3 = 392$, not $72x - 3 = 63$.

Choice C is incorrect. Distributing 7 on the left-hand side of the given equation yields $14x - 21 = 63$, not $2x - 21 = 63$.

Choice D is incorrect. Distributing 7 on the left-hand side of the given equation yields $14x - 21 = 63$, not $2x - 21 = 70$.

▼ 1.51 — 8

The correct answer is 8. Adding x to both sides of the given equation yields $14x = 112$. Dividing both sides of this equation by 14 yields $x = 8$.

▼ 1.52 — B

Choice B is correct. Subtracting 12 from both sides of the given equation yields $k = 324$. Therefore, the solution to the given equation is 324.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.53 — B

Choice B is correct. It's given that Henry uses his $\$ 60.00$ gift card to buy 3 movies for $\$ 7.50$ each. Therefore, Henry spends $3 \$ 7.50$, or $\$ 22.50$, of his $\$ 60.00$ gift card to buy 3 movies. After buying 3 movies with his $\$ 60.00$ gift card, Henry has a gift card balance of $\$ 60.00 - \$ 22.50$, or $\$ 37.50$. It's also given that Henry spends the rest of his gift card balance on renting movies for $\$ 1.50$ each. Therefore, Henry can rent $\frac{\$ 37.50}{\$ 1.50}$, or 25, movies.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.54 — D

Choice D is correct. To solve the equation, use the distributive property to multiply on the right-hand side of the equation which gives w – 10 = 2w + 10. Subtract w from both sides of the equation, which gives –10 = w + 10. Finally, subtract 10 from both sides of the equation, which gives –20 = w.

Choices A and B are incorrect and may result from making sign errors. Choice C is incorrect and may result from incompletely distributing the 2 in the expression 2(w + 5).

▼ 1.55 — A

Choice A is correct. Subtracting 1 from both sides of the given equation, $4x + 1 = 33$, yields $4x = 32$. Therefore, $4x = 32$ has the same solution as the given equation.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.56 — C

Choice C is correct. It's given that $16x + 30 = 190$. Subtracting 30 from each side of this equation yields $16x = 160$. Therefore, the equation $16x = 160$ is equivalent to the given equation and has the same solution.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.